Lab1 Feature Engg.

Source: MLL/Lab1 Feature Engg..ipynb

CELL 1 — CODE | Execution: 56

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder,StandardScaler
```

CELL 2 — CODE | Execution: 87

```
df=pd.read_csv("Titanic-Dataset.csv")
```

CELL 3 — CODE | Execution: 88

```
df
```

OUTPUT

```
     PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                 Name     Sex   Age  SibSp  \
0                             Braund, Mr. Owen Harris    male  22.0      1
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                              Heikkinen, Miss. Laina  female  26.0      0
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                            Allen, Mr. William Henry    male  35.0      0
..                                                ...     ...   ...    ...
886                              Montvila, Rev. Juozas    male  27.0      0
887                       Graham, Miss. Margaret Edith  female  19.0      0
888           Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                              Behr, Mr. Karl Howell    male  26.0      0
890                                Dooley, Mr. Patrick    male  32.0      0

     Parch            Ticket     Fare Cabin Embarked
0        0         A/5 21171   7.2500   NaN        S
1        0          PC 17599  71.2833   C85        C
2        0  STON/O2. 3101282   7.9250   NaN        S
3        0            113803  53.1000  C123        S
4        0            373450   8.0500   NaN        S
..     ...               ...      ...   ...      ...
886      0            211536  13.0000   NaN        S
887      0            112053  30.0000   B42        S
888      2        W./C. 6607  23.4500   NaN        S
889      0            111369  30.0000  C148        C
890      0            370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

CELL 4 — CODE | Execution: 89

```
df.info()
```

OUTPUT

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

CELL 5 — CODE | Execution: 90

```
df.head()
```

OUTPUT

```
   PassengerId  Survived  Pclass  \
0            1         0       3   
1            2         1       1   
2            3         1       3   
3            4         1       1   
4            5         0       3   

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1   
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1   
2                             Heikkinen, Miss. Laina  female  26.0      0   
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1   
4                           Allen, Mr. William Henry    male  35.0      0   

   Parch            Ticket     Fare Cabin Embarked  
0      0         A/5 21171   7.2500   NaN        S  
1      0          PC 17599  71.2833   C85        C  
2      0  STON/O2. 3101282   7.9250   NaN        S  
3      0            113803  53.1000  C123        S  
4      0            373450   8.0500   NaN        S  
```

CELL 6 — CODE | Execution: 91

```
df.tail()
```

OUTPUT

```
     PassengerId  Survived  Pclass                                      Name  \
886          887         0       2                     Montvila, Rev. Juozas   
887          888         1       1              Graham, Miss. Margaret Edith   
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"   
889          890         1       1                     Behr, Mr. Karl Howell   
890          891         0       3                       Dooley, Mr. Patrick   

        Sex   Age  SibSp  Parch      Ticket   Fare Cabin Embarked  
886    male  27.0      0      0      211536  13.00   NaN        S  
887  female  19.0      0      0      112053  30.00   B42        S  
888  female   NaN      1      2  W./C. 6607  23.45   NaN        S  
889    male  26.0      0      0      111369  30.00  C148        C  
890    male  32.0      0      0      370376   7.75   NaN        Q  
```

CELL 7 — CODE | Execution: 92

```
df.describe()
```

OUTPUT

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

CELL 8 — CODE | Execution: 93

```
df.shape
```

OUTPUT

```
(891, 12)
```

CELL 9 — CODE | Execution: 94

```
df.dtypes
```

OUTPUT

```
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

CELL 10 — CODE | Execution: 95

```
df.isnull()
```

OUTPUT

```
     PassengerId  Survived  Pclass   Name    Sex    Age  SibSp  Parch  Ticket  \
0          False     False   False  False  False  False  False  False   False
1          False     False   False  False  False  False  False  False   False
2          False     False   False  False  False  False  False  False   False
3          False     False   False  False  False  False  False  False   False
4          False     False   False  False  False  False  False  False   False
..           ...       ...     ...    ...    ...    ...    ...    ...     ...
886        False     False   False  False  False  False  False  False   False
887        False     False   False  False  False  False  False  False   False
888        False     False   False  False  False  False   True  False   False
889        False     False   False  False  False  False  False  False   False
890        False     False   False  False  False  False  False  False   False

      Fare  Cabin  Embarked
0    False   True     False
1    False  False     False
2    False   True     False
3    False  False     False
4    False   True     False
..     ...    ...       ...
886  False   True     False
887  False  False     False
888  False   True     False
889  False  False     False
890  False   True     False

[891 rows x 12 columns]
```

CELL 11 — CODE | Execution: 96

```
df.notnull()
```

OUTPUT

```
     PassengerId  Survived  Pclass  Name   Sex    Age  SibSp  Parch  Ticket  \
0           True      True    True  True  True   True   True   True    True
1           True      True    True  True  True   True   True   True    True
2           True      True    True  True  True   True   True   True    True
3           True      True    True  True  True   True   True   True    True
4           True      True    True  True  True   True   True   True    True
..           ...       ...     ...   ...   ...    ...    ...    ...     ...
886         True      True    True  True  True   True   True   True    True
887         True      True    True  True  True   True   True   True    True
888         True      True    True  True  True  False   True   True    True
889         True      True    True  True  True   True   True   True    True
890         True      True    True  True  True   True   True   True    True

     Fare  Cabin  Embarked
0    True  False      True
1    True   True      True
2    True  False      True
3    True   True      True
4    True  False      True
..    ...    ...       ...
886  True  False      True
887  True   True      True
888  True  False      True
889  True   True      True
890  True  False      True

[891 rows x 12 columns]
```

CELL 12 — CODE | Execution: 97

```
df.isnull().sum()
```

OUTPUT

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

CELL 13 — CODE | Execution: 98

```
df.fillna(df.mean(numeric_only=True),inplace=True)
df
```

OUTPUT

```
     PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                 Name     Sex        Age  \
0                             Braund, Mr. Owen Harris    male  22.000000
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.000000
2                             Heikkinen, Miss. Laina  female  26.000000
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.000000
4                           Allen, Mr. William Henry    male  35.000000
..                                               ...     ...        ...
886                            Montvila, Rev. Juozas    male  27.000000
887                     Graham, Miss. Margaret Edith  female  19.000000
888         Johnston, Miss. Catherine Helen "Carrie"  female  29.699118
889                            Behr, Mr. Karl Howell    male  26.000000
890                              Dooley, Mr. Patrick    male  32.000000

     SibSp  Parch            Ticket     Fare Cabin Embarked
0        1      0         A/5 21171   7.2500   NaN        S
1        1      0          PC 17599  71.2833   C85        C
2        0      0  STON/O2. 3101282   7.9250   NaN        S
3        1      0            113803  53.1000  C123        S
4        0      0            373450   8.0500   NaN        S
..     ...    ...               ...      ...   ...      ...
886      0      0            211536  13.0000   NaN        S
887      0      0            112053  30.0000   B42        S
888      1      2        W./C. 6607  23.4500   NaN        S
889      0      0            111369  30.0000  C148        C
890      0      0            370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

CELL 14 — CODE | Execution: 99

```
label=LabelEncoder()
df['Name']=label.fit_transform(df['Name'])
df
```

OUTPUT

```
     PassengerId  Survived  Pclass  Name     Sex        Age  SibSp  Parch  \
0              1         0       3   108    male  22.000000      1      0
1              2         1       1   190  female  38.000000      1      0
2              3         1       3   353  female  26.000000      0      0
3              4         1       1   272  female  35.000000      1      0
4              5         0       3    15    male  35.000000      0      0
..           ...       ...     ...   ...     ...        ...    ...    ...
886          887         0       2   548    male  27.000000      0      0
887          888         1       1   303  female  19.000000      0      0
888          889         0       3   413  female  29.699118      1      2
889          890         1       1    81    male  26.000000      0      0
890          891         0       3   220    male  32.000000      0      0

               Ticket     Fare Cabin Embarked
0           A/5 21171   7.2500   NaN        S
1            PC 17599  71.2833   C85        C
2    STON/O2. 3101282   7.9250   NaN        S
3              113803  53.1000  C123        S
4              373450   8.0500   NaN        S
..                ...      ...   ...      ...
886            211536  13.0000   NaN        S
887            112053  30.0000   B42        S
888        W./C. 6607  23.4500   NaN        S
889            111369  30.0000  C148        C
890            370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

CELL 15 — CODE | Execution: 100

```
Scaler=StandardScaler()
num_value= ['Fare']
df[num_value]=Scaler.fit_transform(df[num_value])
df
```

OUTPUT

```
     PassengerId  Survived  Pclass  Name     Sex        Age  SibSp  Parch  \
0              1         0       3   108    male  22.000000      1      0
1              2         1       1   190  female  38.000000      1      0
2              3         1       3   353  female  26.000000      0      0
3              4         1       1   272  female  35.000000      1      0
4              5         0       3    15    male  35.000000      0      0
..           ...       ...     ...   ...     ...        ...    ...    ...
886          887         0       2   548    male  27.000000      0      0
887          888         1       1   303  female  19.000000      0      0
888          889         0       3   413  female  29.699118      1      2
889          890         1       1    81    male  26.000000      0      0
890          891         0       3   220    male  32.000000      0      0

               Ticket      Fare Cabin Embarked
0           A/5 21171 -0.502445   NaN        S
1            PC 17599  0.786845   C85        C
2    STON/O2. 3101282 -0.488854   NaN        S
3              113803  0.420730  C123        S
4              373450 -0.486337   NaN        S
..                ...       ...   ...      ...
886            211536 -0.386671   NaN        S
887            112053 -0.044381   B42        S
888        W./C. 6607 -0.176263   NaN        S
889            111369 -0.044381  C148        C
890            370376 -0.492378   NaN        Q

[891 rows x 12 columns]
```

CELL 16 — CODE | Execution: 101

```
df = df.drop('Cabin', axis=1)
```

CELL 17 — CODE | Execution: 102

```
df
```

OUTPUT

```
     PassengerId  Survived  Pclass  Name     Sex        Age  SibSp  Parch  \
0              1         0       3   108    male  22.000000      1      0
1              2         1       1   190  female  38.000000      1      0
2              3         1       3   353  female  26.000000      0      0
3              4         1       1   272  female  35.000000      1      0
4              5         0       3    15    male  35.000000      0      0
..           ...       ...     ...   ...     ...        ...    ...    ...
886          887         0       2   548    male  27.000000      0      0
887          888         1       1   303  female  19.000000      0      0
888          889         0       3   413  female  29.699118      1      2
889          890         1       1    81    male  26.000000      0      0
890          891         0       3   220    male  32.000000      0      0

               Ticket      Fare Embarked
0           A/5 21171 -0.502445        S
1            PC 17599  0.786845        C
2    STON/O2. 3101282 -0.488854        S
3              113803  0.420730        S
4              373450 -0.486337        S
..                ...       ...      ...
886            211536 -0.386671        S
887            112053 -0.044381        S
888        W./C. 6607 -0.176263        S
889            111369 -0.044381        C
890            370376 -0.492378        Q

[891 rows x 11 columns]
```

CELL 18 — CODE

(Empty cell)